2 Newton's Laws

Chapter Outline

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CENGAGE

2.1 Force

The key concept in Newtonian mechanics is **force**.

DEFINITION

Force:

A push or a pull acting on a body. Force usually causes some distortion of the body, a change in its velocity, or both. Force is a vector.

Force is the agent for *a change in motion*

2.1 Force

- We exert forces on objects in dozens of everyday situations, such as in pushing or pulling a door open, lifting a book, pulling out a drawer, and throwing a ball.

- Many machines we use are designed to exert forces. Cranes, hoists, jacks, and vises are all examples of such machines.

- Automobiles and other propelled vehicles function by causing forces to act on them.

2.1 Force

The conversion factors between the primary units of force in the two measurement systems are:

$$1\ \mathbf{N} = 0.225\ \mathbf{lb} \rightarrow \mathbf{1\ lb} = 4.45\ \mathbf{N}$$

The following table lists the common force units of measure in the *metric and English systems* and their abbreviations.

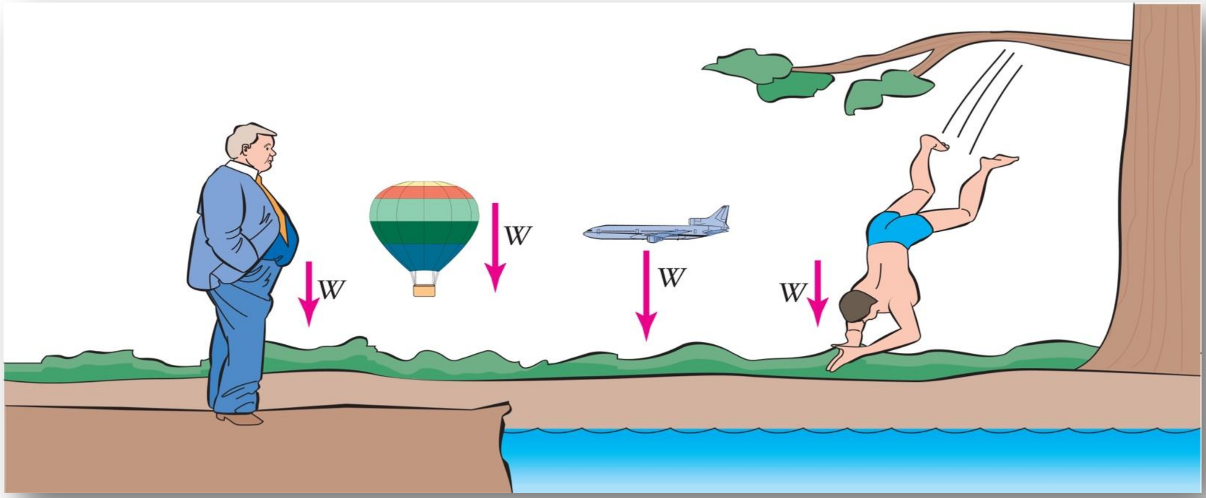
Physical Quantity	**Metric Units**	**English Units**
Force (*F*)	newton (N)	pound (lb)
	dyne	ounce (oz)
	metric ton	ton

2.1 Force: Weight

The most common force in our lives is **weight**.

DEFINITION

Weight:
The force of gravity acting on a body. Symbolized by W.



2.1 Force: Weight

The weight of an object depends on two things:

I. the amount of matter comprising the object (its mass)

II. The acceleration due to gravity force, meaning that a body's weight depends on where it is.

2.1 Force: Weight

- The weight of an object on the Moon is about one-sixth the weight it would have on Earth because the gravitational pull of the Moon is less than Earth's.

- Even on Earth, the weight of an object varies slightly with location. A 190-pound person would weigh about 1 pound less at the equator than at the North or South Pole.

2.1 Force: Friction

DEFINITION

Friction: A force of resistance to relative motion between two bodies or substances in physical contact.

Friction is at work when:

- A chair slides across a floor.
- Brakes keep a car from rolling down a hill.
- Air resistance slows down a baseball.
- A boat glides over the water.

2.1 Force: Friction

There are two types of friction

- **Static friction** is the resistance you experience when you try to *start* sliding an object.
  - It is static because there is no relative motion between the surfaces.
- **Kinetic friction** is the resistance you experience *while* sliding an object.
  - It is kinetic since there is relative motion.

2.2 Newton's First Law of Motion

Newton's laws make no reference to specific types of forces. They are general statements that apply whether a force is caused by.

LAWS

Newton's First Law of Motion: An object will remain at rest or in uniform motion with constant velocity unless acted on by a net external force.

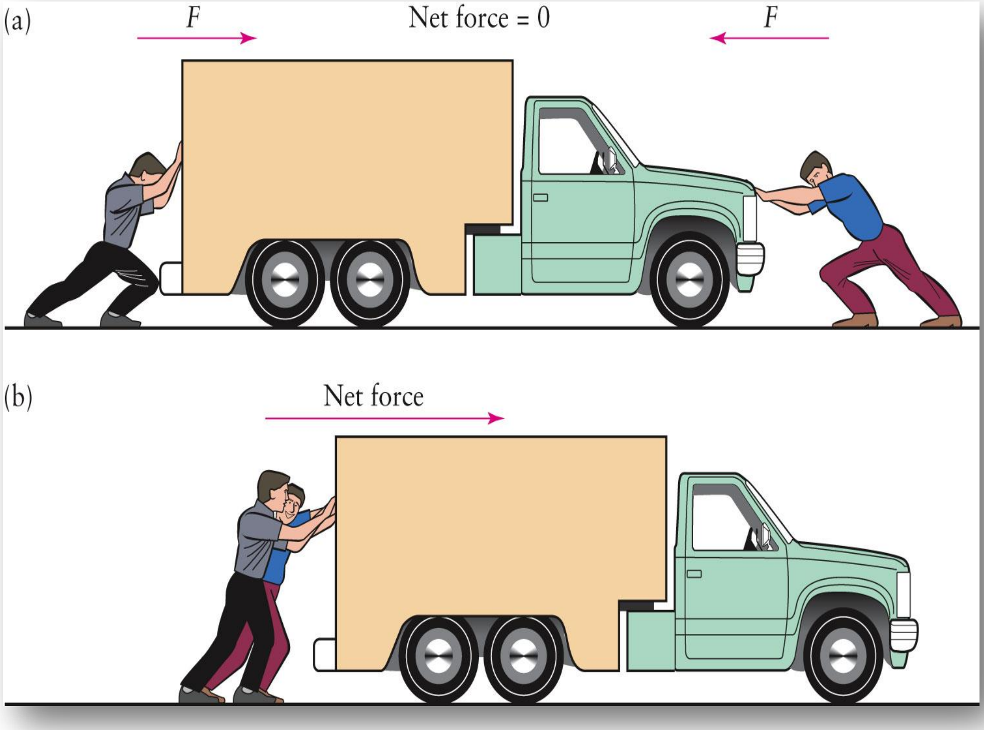
2.2 Newton's First Law of Motion

- An **external force** is a force applied to the object from some other object.

  Weight is an external force because it is caused by something outside an object (Earth, for example)

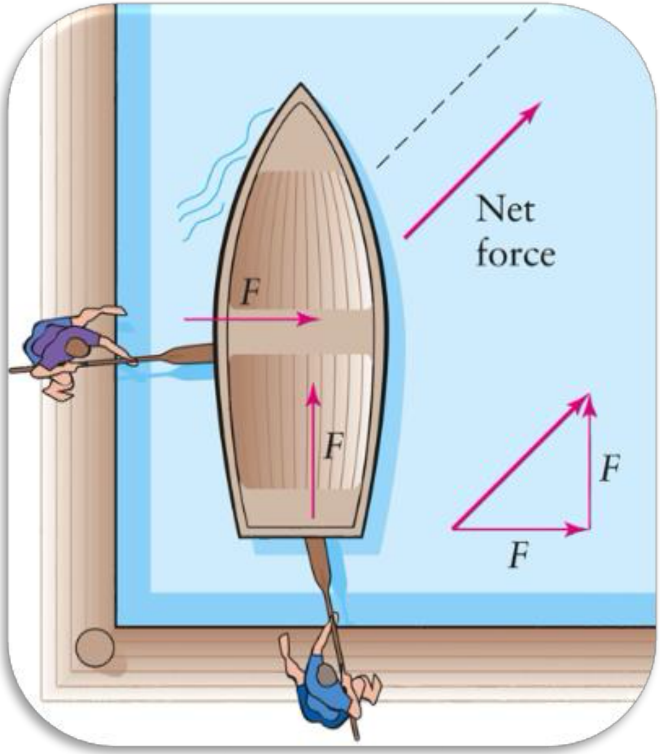
- The ***net* force** is the vector sum of all the external forces acting on the body.

2.2 Newton's First Law of Motion



If two forces act in the same direction, the net force is the sum of the two.

2.2 Newton's First Law of Motion



If two forces act in different directions, the net force is found by adding the two vectors together.

2.2 Newton's First Law of Motion

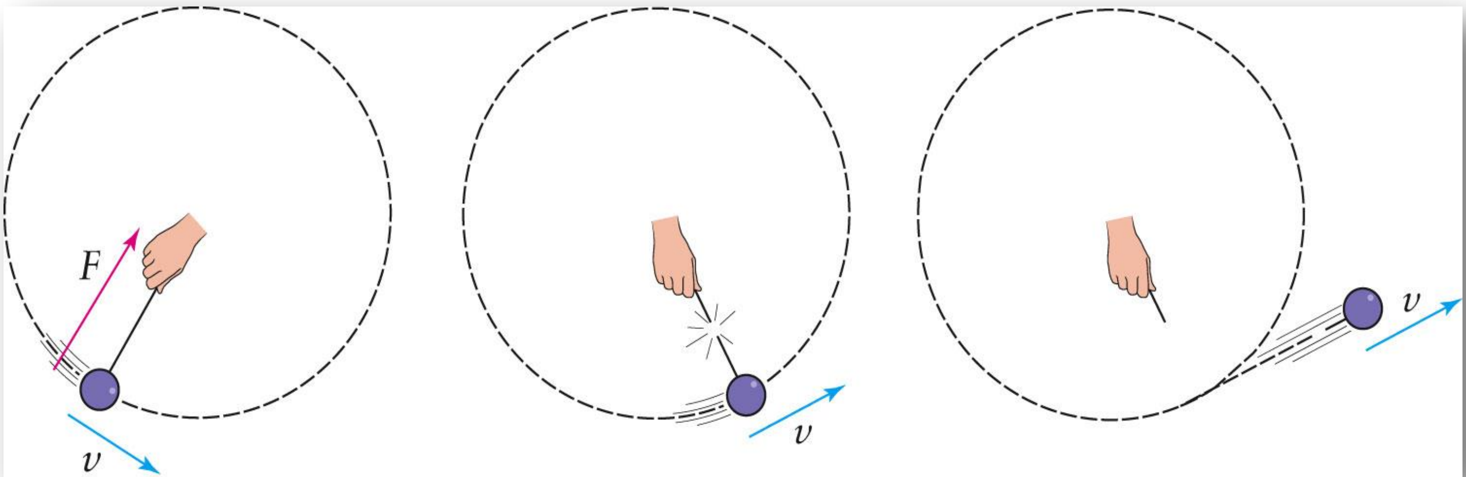
Upon first reading, Newton's first law does not seem to be terribly profound. Obviously, an object will remain stationary unless a net force causes it to move. But the law also states that anything that is already moving with a certain velocity will not speed up, slow down, or change direction unless a net force acts on it.

Newton's first law implies that a force is required only to ***change*** the state of motion.

2.2 Newton's First Law of Motion: Centripetal Force

The first law tells us that a force is required to produce any acceleration. The centripetal acceleration inherent in circular motion requires a force called the **centripetal force**.

The centripetal force on an object is directed toward the center of the object's path



2.3 Mass

The concept of inertia is embodied in the physical quantity **mass**.

DEFINITION

Mass:

A measure of an object’s resistance to acceleration. A measure of the quantity of matter in an object.

2.3 Mass

The following table lists the mass units of measure, in both the *metric and English systems* and their abbreviations.

Physical Quantity	Metric Units	English Units
Mass (*m*)	kilogram (kg)	slug
	gram (g)	

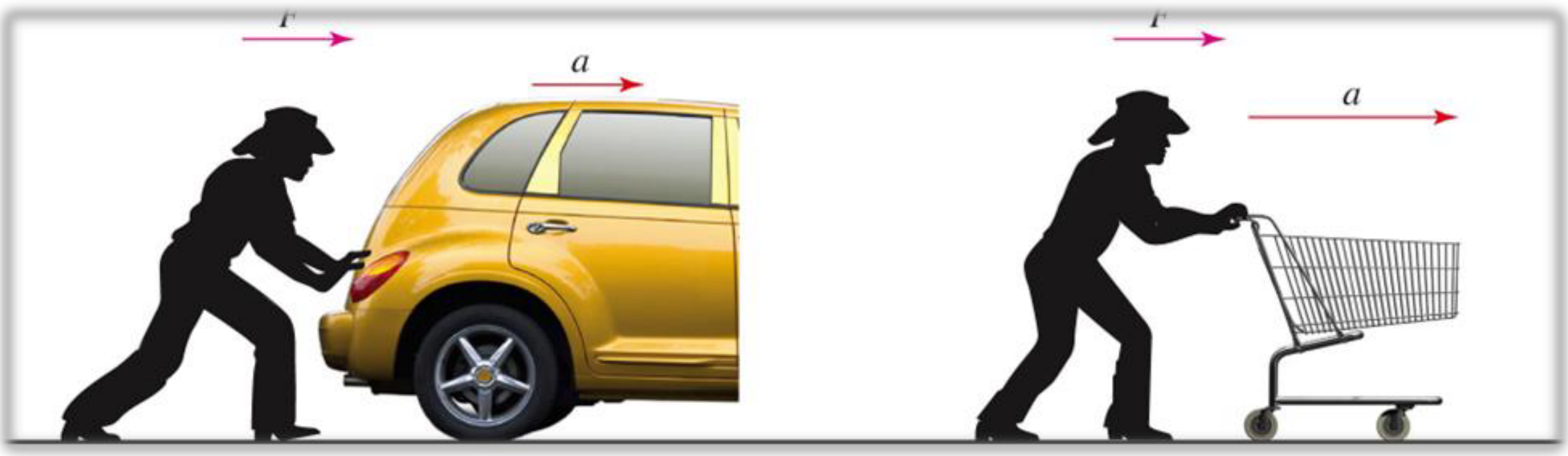
2.3 Mass

The actual mass of a given object depends on its size (volume) and its composition. A rock has more mass than a pebble because of its size; it has more mass than an otherwise identical piece of Styrofoam because of its composition.

Mass is *not* the same as weight. Mass and weight are related to each other in that the weight of an object is proportional to its mass. But weight is a force that arises from a gravitational interaction. Mass is an intrinsic property of matter that does not depend on any external phenomenon.

2.3 Mass

- An object of little mass requires little force to accelerate it.
- A massive object requires a much larger force to give it the same acceleration.



2.3 Mass

- The weight of an object depends on which planet you measure the weight.
- The mass of the object is independent of the planet.

2.4 Newton's 2^{nd} law of motion: Force and Acceleration

LAWS

Newton's Second Law of Motion: An object is accelerated whenever a net external force acts on it. The net force equals the object's mass times its acceleration:

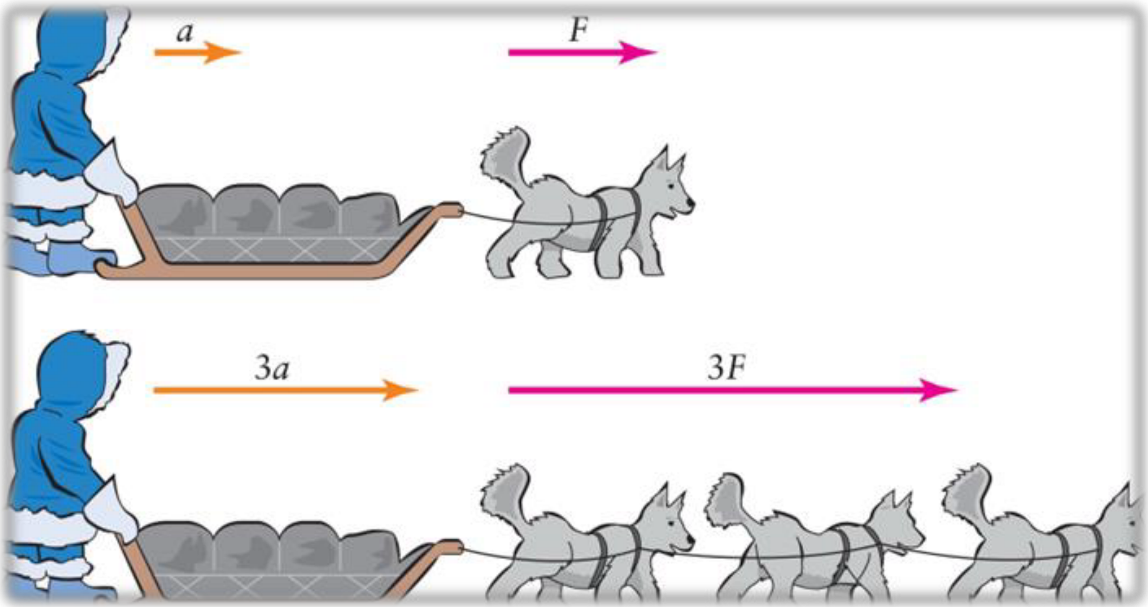
$$F = ma$$

Newton's Second Law expresses the exact relationship between force and acceleration.

2.4 Newton's 2nd law of motion: Force and Acceleration

- For a given body, a larger force will cause a proportionally larger acceleration.

- Force and acceleration both are vectors. The direction of the acceleration of an object is the same as the direction of the net force acting on it.



2.4 Newton's 2nd law of motion: Force and Acceleration

The unit of measure of force, the newton, a derived unit, is established by this law:

$$F = ma$$

$$1\ \text{N} = (1\ \text{kg})\left(1\ \frac{\text{m}}{\text{s}^2}\right) = 1\ \frac{\text{kg}\cdot\text{m}}{\text{s}^2}$$

One newton is the force required to give a 1-kilogram mass an acceleration of 1 m/s.

Example 2.1

An airplane with a mass of 2,000 kilograms is observed to be accelerating at a rate of 4 m/s^2. What is the net force acting on it?

$$F = ma = 2{,}000\ \mathrm{kg} \times 4\ \mathrm{m/s^2} = 8{,}000\ \mathrm{N}$$

2.4 Newton's 2nd law of motion: Force and Acceleration

Alternate statement of Newton's second law:

$$a = \frac{F}{m}$$

$$a = \frac{8{,}000\ \mathrm{N}}{2{,}000\ \mathrm{kg}} = \frac{4\ \mathrm{N}}{1\ \mathrm{kg}} = 4\ \mathrm{m/s^2}$$

Example 2.2

An automobile manufacturer decides to build a car that can accelerate uniformly from 0 to 60 mph in 10 s. In metric units, this is from 0 to 27 m/s. The car's mass is to be about 1,000 kilograms.

What is the force required?

$$a = \frac{\Delta v}{\Delta t} = \frac{27\ m/s - 0\ m/s}{10\ s} = 2.7\ \mathrm{m/s^2}$$

$$F = ma = 1{,}000\ \mathrm{kg} \times 2.7\ \mathrm{m/s^2} = 2{,}700\ \mathrm{N} = 607.5\ \mathrm{lb}$$

2.4 Newton's 2nd law of motion: Force and Acceleration

Newton's second law establishes the relationship between mass and weight.

A falling object accelerates due to the Earth's gravity at

$$g = 9.8\ \mathrm{m/s^2}$$

By the second law, the size of the gravitational force needed to cause this acceleration is:

$$F = ma = mg$$
$$F = ma \rightarrow W = mg$$

We call this force the object's ***weight***.

2.4 Newton's 2nd law of motion: Force and Acceleration

On Earth, where the acceleration from gravity is 9.8 m/s^2, the weight of an object is:

$$W = m \times 9.8\ \mathrm{m/s^2} \text{(on Earth, } m \text{ in kg, } W \text{ in N)}$$

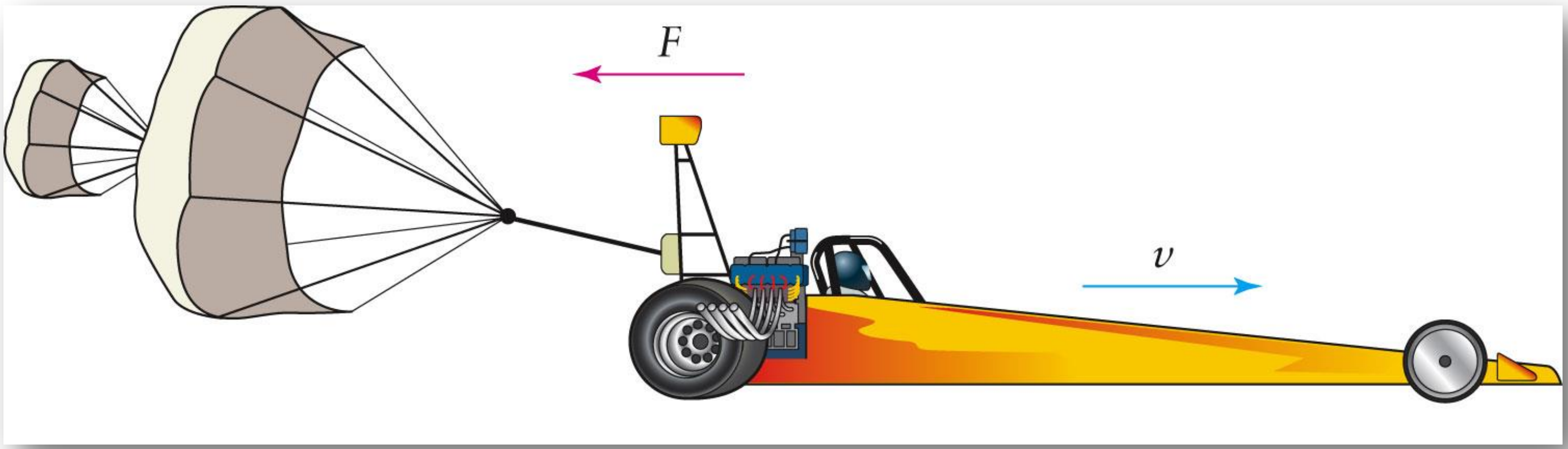
On the Moon, the acceleration from gravity is 1.6 m/s^2. Therefore,

$$W = m \times 1.6 \text{ (on the Moon, } m \text{ in kg, } W \text{ in N)}$$

2.4 Newton's 2nd law of motion: Force and Acceleration

Forces that cause deceleration or centripetal acceleration also follow the second law.

To slow down a moving object, a force must act in the direction opposite the object's velocity.

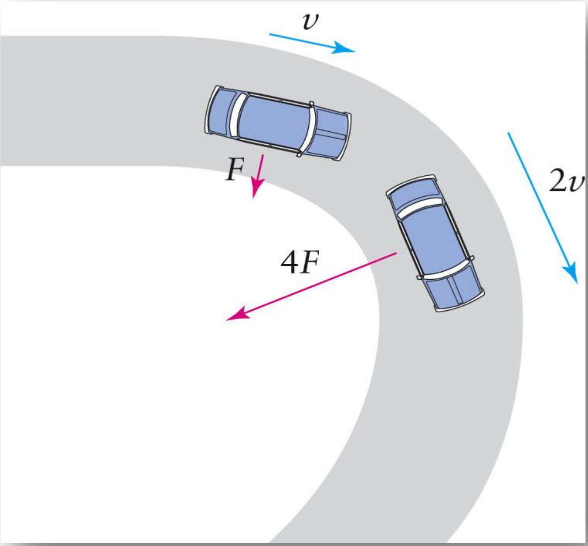


2.4 Newton's 2nd law of motion: Force and Acceleration

A force acting sideways on a moving body causes a centripetal acceleration.

The object moves along a circular path as long as the net force remains perpendicular to the velocity.

The size of the centripetal force required is:



Centripetal Force:

$F = ma$ **and because** $a = \dfrac{v^2}{r} \rightarrow$

$$F = \frac{mv^2}{r} \quad \text{(centripetal force)}$$

Example 2.3

A car is going 10 m/s around a curve with a radius of 20 meters. If the car's mass is 1,000 kilograms, what is the centripetal force that acts on it?

$$F = \frac{mv^2}{r} = \frac{1{,}000\ \mathbf{kg} \times (10\ \mathbf{m/s})^2}{20\ \mathbf{m}}$$
$$= 5{,}000\ \mathbf{N} = 1{,}120\ \mathbf{lb}$$

$$F = ma = 1{,}000\ \mathbf{kg} \times 5.0\ \mathbf{m/s}^2 = 5{,}000\ \mathbf{N}$$

2.4 Newton's 2^{nd} law of motion: Force and Acceleration

The form of the second law introduced here, $F = ma$, is not the version originally formulated by Newton, but it is the most useful one for our purposes. It applies only when the mass of the object doesn't change. This is almost always the case, but there are important exceptions.

2.4 Newton's 2nd law of motion: The International System of Units (SI)

Physical Quantity	**SI Unit**
Distance (d)	meter (m)
Area (A)	square meter (m^2)
Volume (V)	cubic meter (m^3)
Time (t)	second (s)
Frequency (f)	hertz (Hz)
Speed and velocity (v)	meter per second (m/s)
Acceleration (a)	meter per second squared (m/s^2)
Force (F) and weight (W)	newton (N)
Mass (m)	kilogram (kg)

2.5 Examples: Different Forces, Different Motions

Previously, we considered three simple states of motion, zero velocity, constant velocity, and constant acceleration. Now, we will examine how force is involved in these cases and then look at some other types of motion.

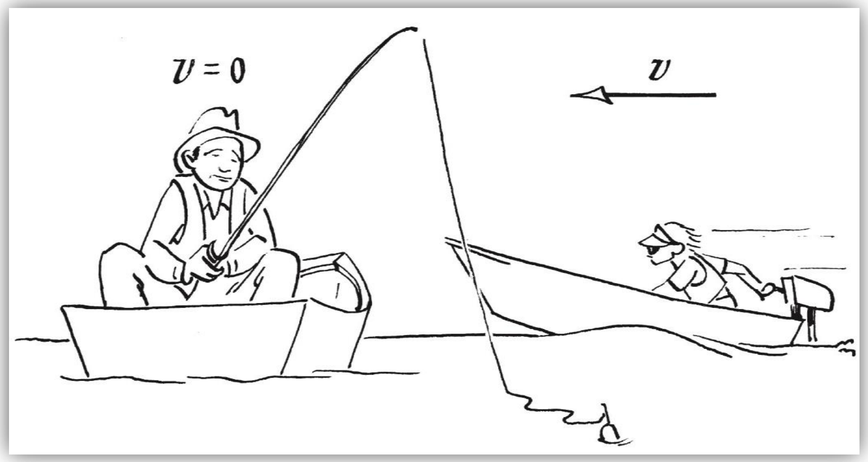
Here, the particular cause of the force is not important, our concern is the relationship between the force—both its magnitude and direction—and the motion.

2.5 Examples: Different Forces, Different Motions

- The acceleration of a stationary object or of one moving with uniform motion (constant velocity) is zero. By the second law, this means that the net force is also zero.

- When the net force on a body is zero, we say that it is in ***equilibrium***, whether it is stationary or moving with

constant velocity.



2.5 Examples: Different Forces, Different Motions

- In uniform acceleration, the second law implies that a constant force acts to produce a ***constant acceleration*****.**

- A steady net force acting in a fixed direction will make an object move with a constant acceleration.

- A freely falling body is a good example: the force (weight) is constant and always acts in a downward direction.

- Free-fall is a type of motion in which the only force acting on the object is the object's weight. There is no friction, no air resistance, etc.

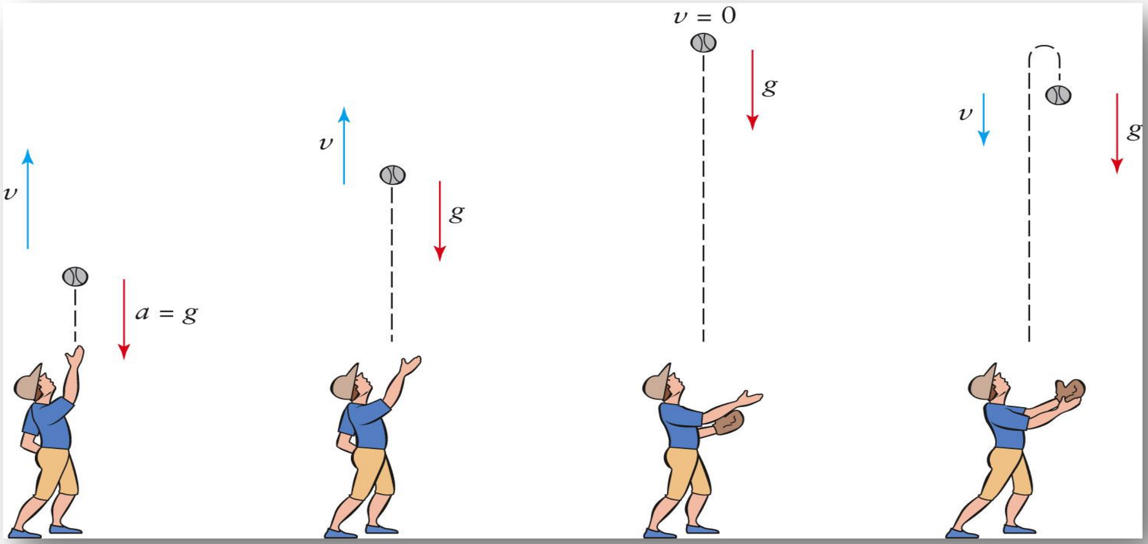
2.5 Examples: Different Forces, Different Motions

A net force that acts opposite to the direction of motion of an object will cause it to slow down.

If the force continues to act, the object will come to a stop at an instant and then accelerate in the direction of the force—that is, opposite its original direction of motion.

2.5 Examples: Different Forces, Different Motions

- A ball thrown upward will slow-down because gravity is "pulling" down on the ball.
- It is accelerating downward the whole time.
- on the way up, its speed decreases until it reaches its highest point.
- There, it has zero speed at an instant and then begins to fall with increasing speed.

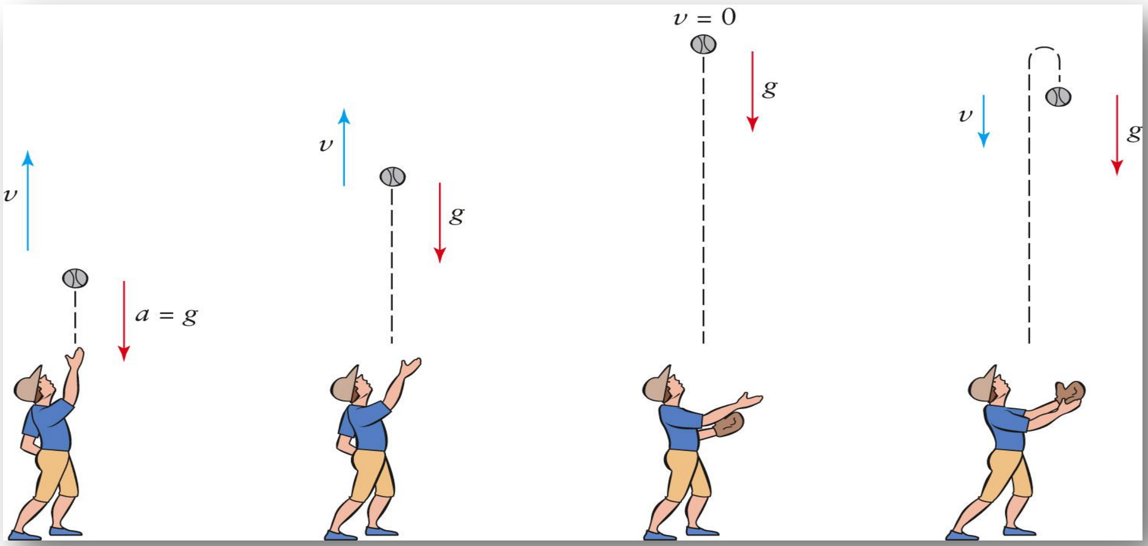


2.5 Examples: Different Forces, Different Motions

- Its acceleration is always *g*, even at the instant that it is stopped.

- It will then fall back to the Earth due to gravity.

- The time it takes to go up is the same amount of time for it to fall.

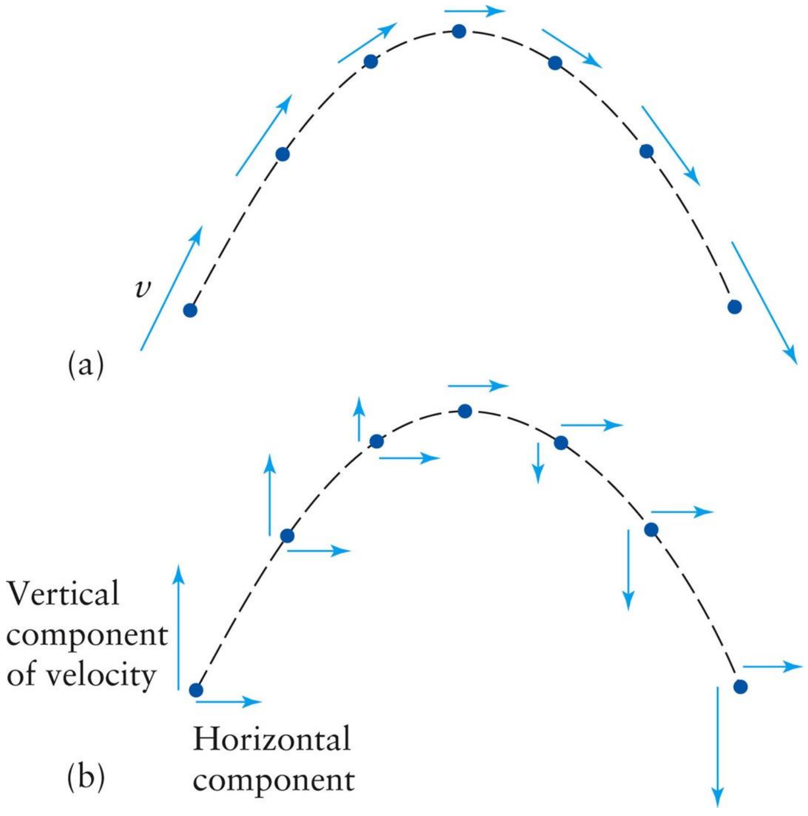


2.5 Examples: Different Forces, Different Motions: Projectile Motion

- Projectile motion is a composite of horizontal and vertical motions.

- The path that a projectile takes has a characteristic arc shape.

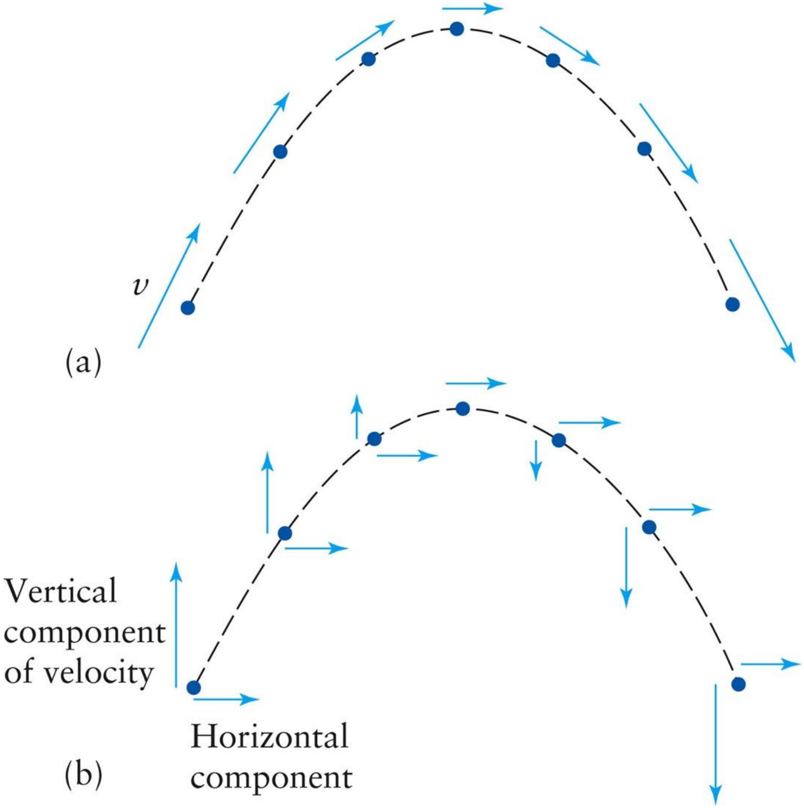
- This shape, an important one in mathematics, is called a *parabola.*



2.5 Examples: Different Forces, Different Motions: Projectile Motion

- The key to understanding projectile motion is to realize that the vertical force of gravity has no effect on the horizontal motion.

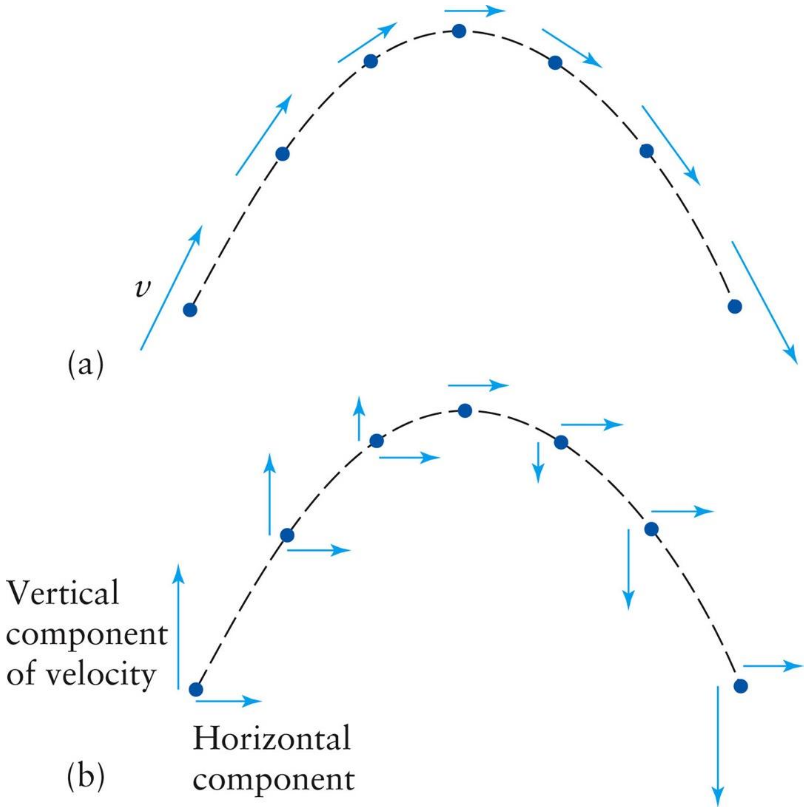
- An object initially moving horizontally has the same downward acceleration *g* as an object that is simply dropped.



2.5 Examples: Different Forces, Different Motions: Projectile Motion

- As an object moves along its path, the vertical component of its velocity decreases to zero (at the highest point of the arc) and then increases downward.

- The horizontal component of its velocity stays constant.



2.5 Examples: Different Forces, Different Motions: Projectile Motion

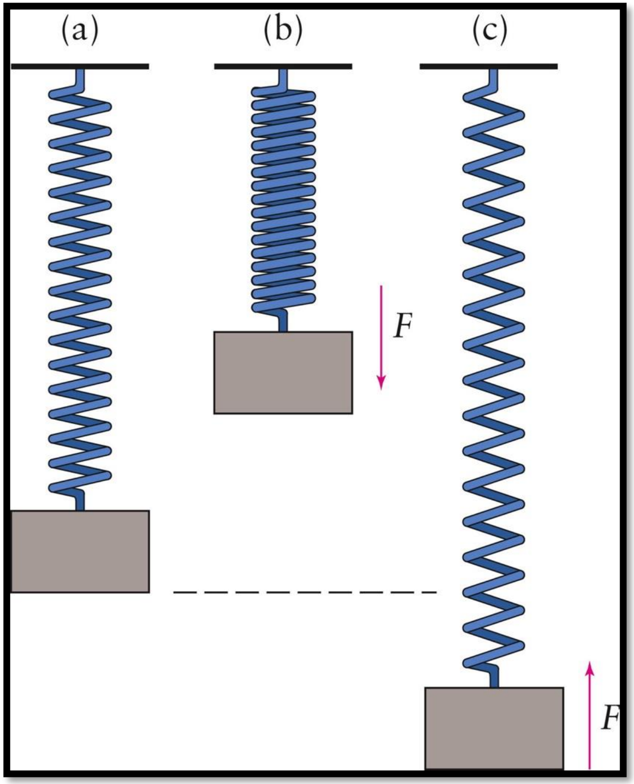
Ignoring air resistance, a projectile moves horizontally with constant speed and vertically with constant downward acceleration.

2.5 Examples: Different Forces, Different Motions: Simple Harmonic Motion

- Imagine a block attached to a spring.

- When it is not moving, the block is in equilibrium, and the net force on it is zero.

- If we stretch the spring and release it, the block moves in a regular, periodic fashion.

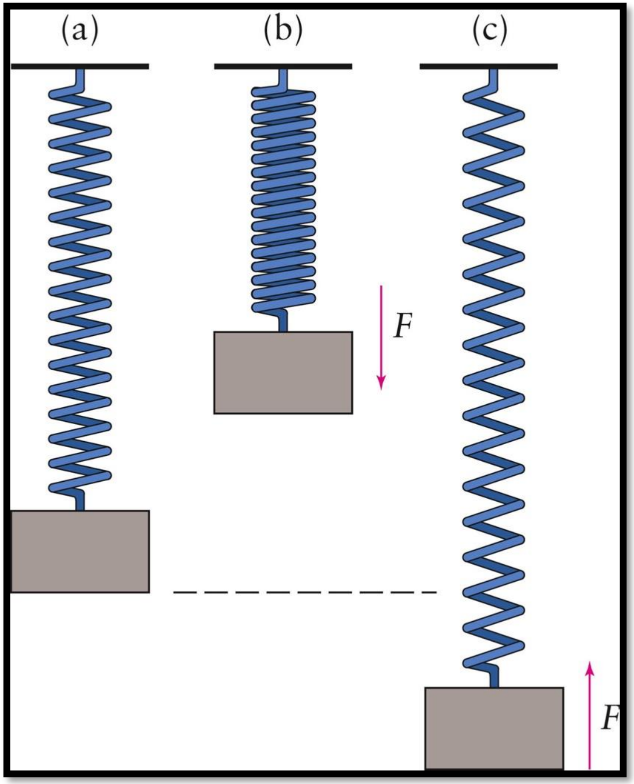


2.5 Examples: Different Forces, Different Motions: Simple Harmonic Motion

- This type of motion is called **simple harmonic motion**.

- If you lift the block up a bit and then release it, it will experience a net force downward, toward its original (rest) position.

- if you pull it down a bit below its rest position and release it. Then the net force on it will be up, again toward its original position.



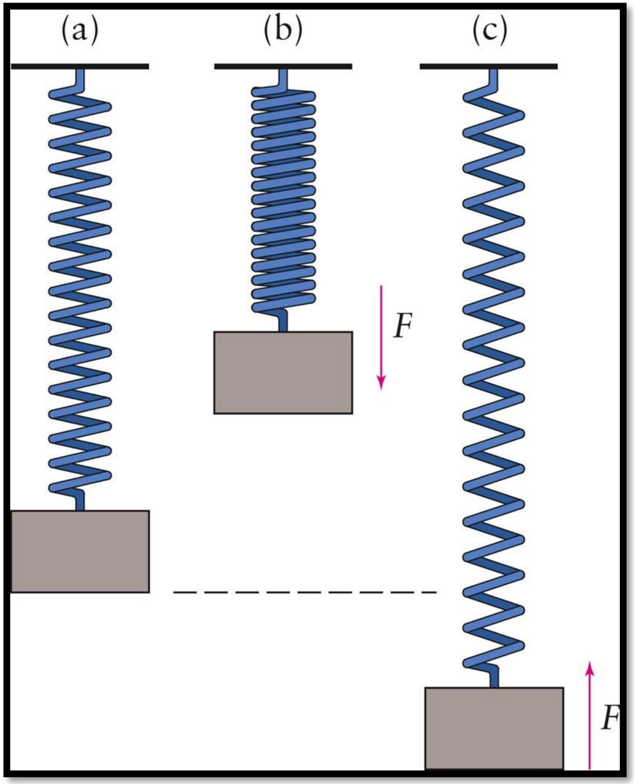
2.5 Examples: Different Forces, Different Motions: Simple Harmonic Motion

- The force that cause this motion is not constant

- Such a force is called a *restoring force* because it acts to restore the system to the original configuration.

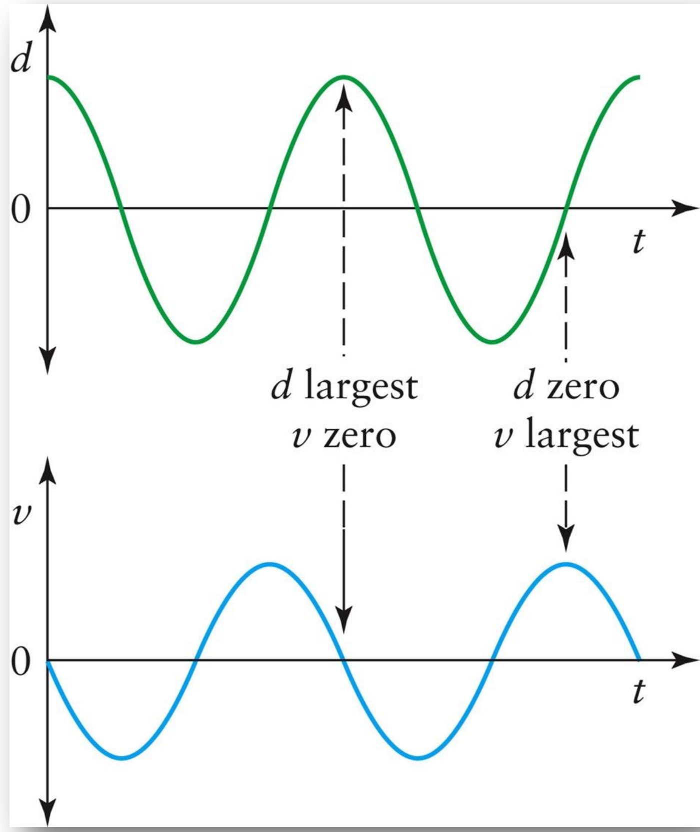
- In this case, the net force is proportional to the displacement from the rest position.

- The farther the block is displaced, the greater the force acting to move it back.



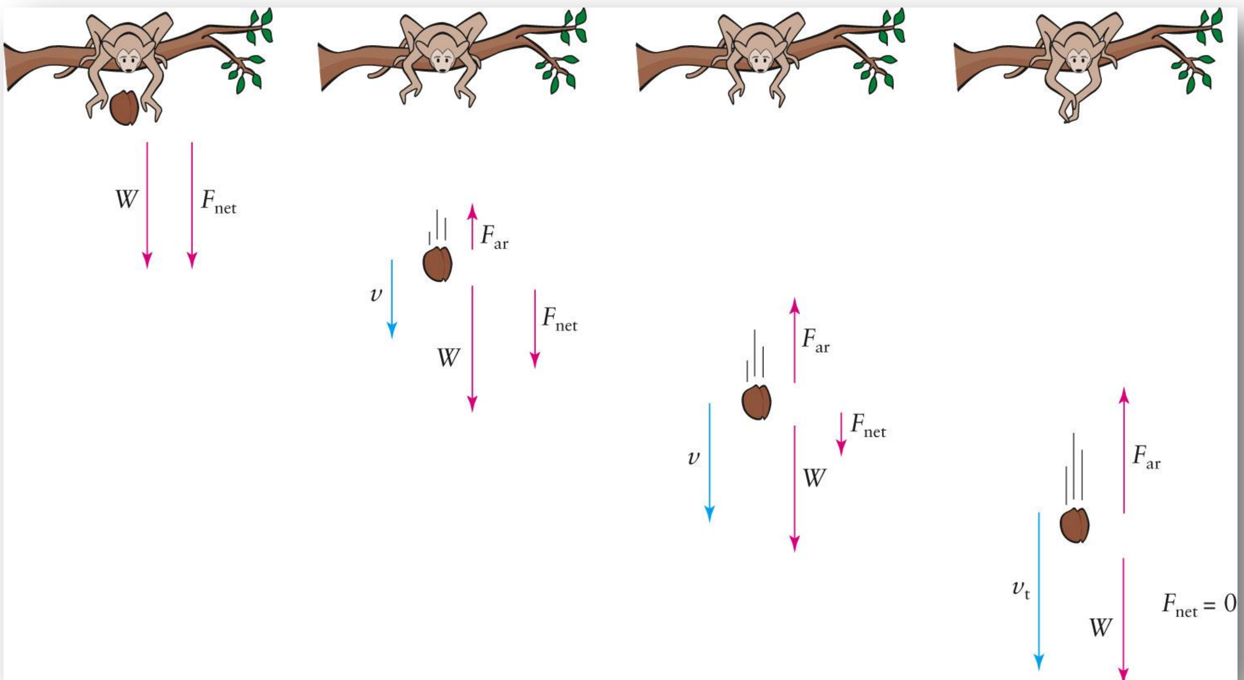
2.5 Examples: Different Forces, Different Motions: Simple Harmonic Motion

Notice that the speed is minimal when the block is farthest from its rest position.



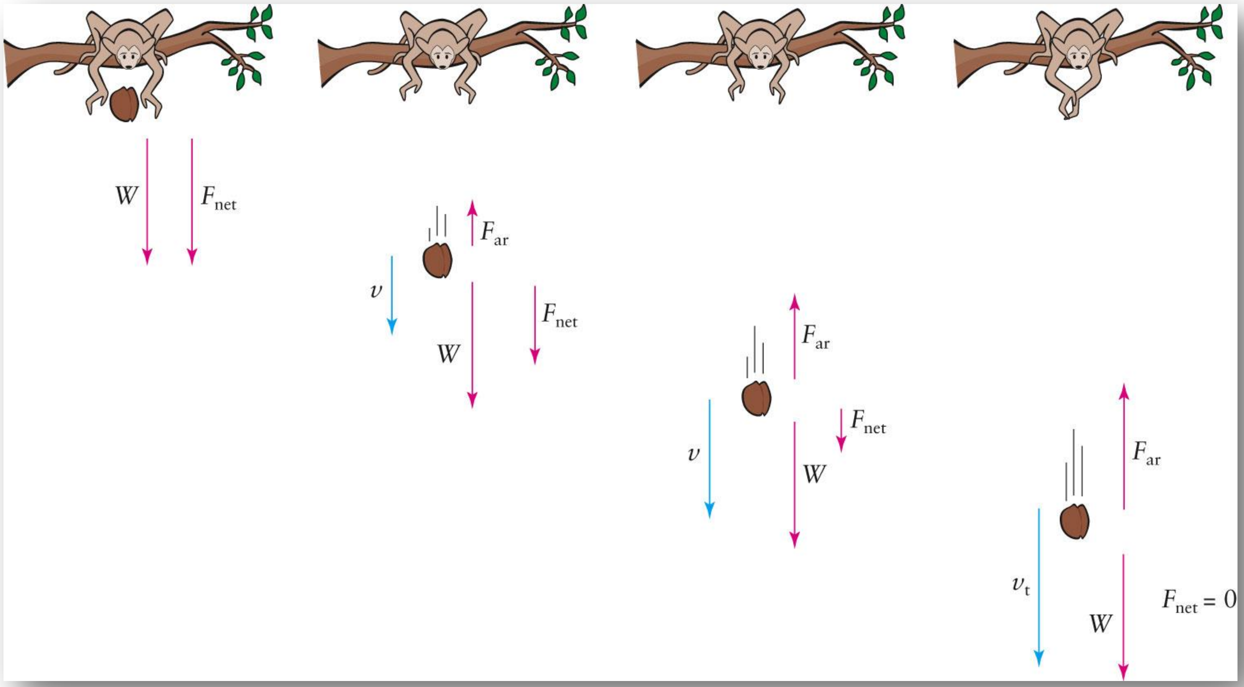
2.5 Examples: Different Forces, Different Motions: Falling Body with Air Resistance

- The force of air resistance is in the opposite direction of the object's velocity and will cause the object to slow down if no other force opposes it.
- The faster an object goes, the larger the force of air resistance.



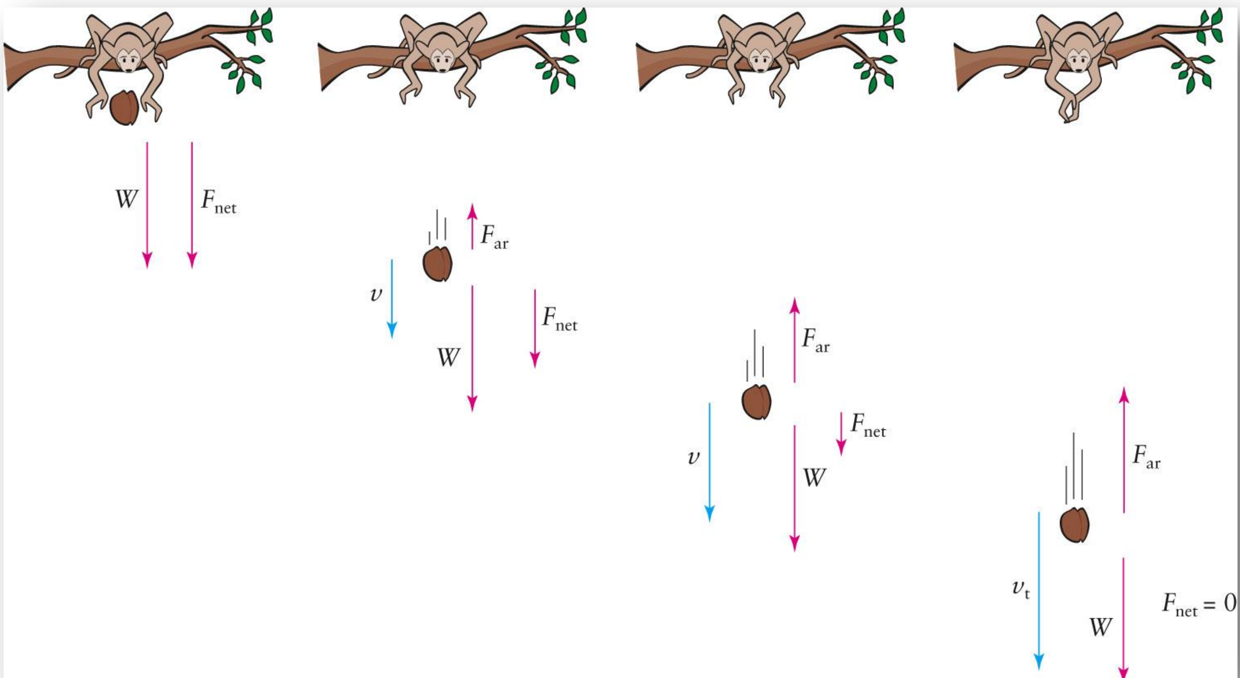
2.5 Examples: Different Forces, Different Motions: Falling Body with Air Resistance

- As a body falls through the air, the force of air resistance grows as the speed increases and eventually affects the motion.
- This increasing force acts opposite to the downward force of gravity, so the net force decreases.



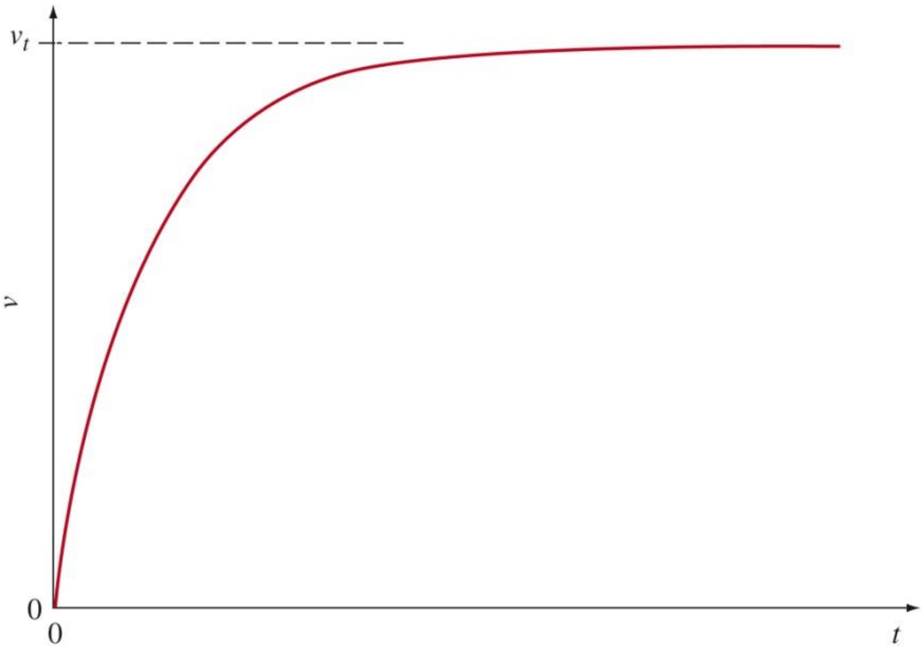
2.5 Examples: Different Forces, Different Motions: Falling Body with Air Resistance

This continues as the body gains speed until the force of air resistance acting upward equals the weight force acting downward.



2.5 Examples: Different Forces, Different Motions: Falling Body with Air Resistance

At this point, the net force is zero, so the acceleration vanishes, and the speed remains constant from then on. This speed is called the ***terminal speed* of the body**.



2.5 Examples: Different Forces, Different Motions

Nature of Net Force	**Description of Motion**
Zero net force	Constant velocity; stationary or motion in straight line with constant speed
Constant net force Force parallel to velocity Force opposite to velocity Force perpendicular to velocity	Constant acceleration Motion in a straight line with increasing speed Motion in a straight line with decreasing speed Motion in a circle: radius depends on speed and force
Restoring force proportional to displacement	Simple harmonic motion (oscillation)
Net force decreases as speed increases	Acceleration decreases: velocity reaches a constant value

2.6 Newton's Third Law of Motion

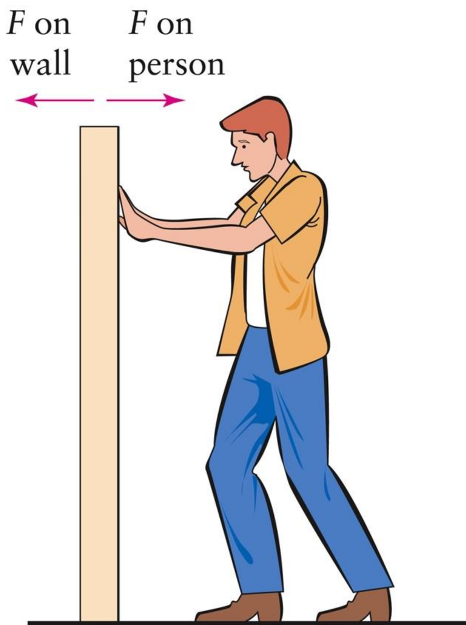
LAWS

Newton's Third Law of Motion:
states that whenever one body exerts a force on a second one, the second exerts an equal and opposite force on the first.

forces always occur in pairs

If object A causes a force on object B, then object B exerts an equal force in the opposite direction on A.

$$F_{\text{B on A}} = -F_{\text{A on B}}$$



2.6 Newton's Third Law of Motion

Rockets and jet aircrafts are propelled by ejecting combustion gases at a high speed. The engine exert a force that ejects the gases, and the gases exert an equal and opposite force on the rocket or jet.

2.6 Newton's Third Law of Motion

Like Newton's first law, the third law is mainly conceptual rather than mathematical.

It emphasizes that forces arise only during interactions between two or more things. By

Example 2.5

During an EVA ("extra-vehicular activity") to repair a camera on board the Hubble Space Telescope, an astronaut exerts a force of 125 N on the spacecraft. (a) How much force, if any, does the HST exert on the astronaut during this interaction?

$$F_{\text{astronaut } \mathbf{on\ HST}} = -F_{\mathbf{HST\ on\ astronaut}}$$
$$\rightarrow F_{\mathbf{HST\ on}\text{ astronaut}} = -125\ \mathbf{N}$$

Example 2.5, continued

(b) If the astronaut (plus life-support equipment) has a mass of 85 kg and the mass of the spacecraft is 11,600 kg, find the resultant acceleration of each body just after the interaction.

$$F_{\mathbf{HST}} = m_{\text{HST}} \times a_{\text{HST}}$$
$$125\ \mathbf{N} = 11{,}600\ \mathbf{kg} \times a_{\text{HST}}$$
$$a_{\text{HST}} = (125\ \mathbf{N}) \div 11{,}600\ \mathbf{kg} = 0.0108\ \mathbf{m/s}^2$$

$$F_{\text{astro}} = m_{\text{astro}} \times a_{\text{astro}}$$
$$125\ \mathbf{N} = 85\ \mathbf{kg} \times a_{\text{astro}}$$
$$a_{\text{astro}} = (125\ \mathbf{N}) \div 85\ \mathbf{kg}$$
$$= 1.52\ \mathbf{m/s}^2$$

2.7 The Law of Universal Gravitation

- Newton's fourth major contribution to the study of mechanics is not a law of motion but a law relating to gravity.

- The concept of ***universal gravitation*** means *that* gravity acts everywhere and on all things.

- The force that Earth exerts on objects near its surface—weight—is just one example of universal gravitation.

2.7 The Law of Universal Gravitation

LAWS

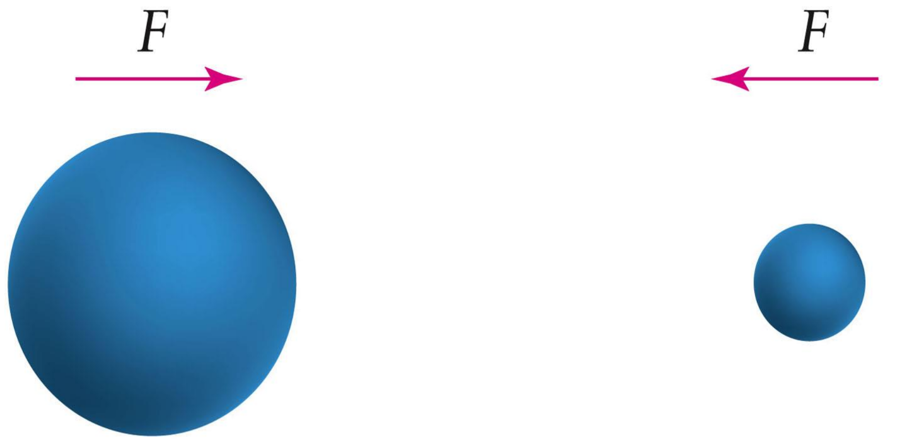
Newton's Law of Universal Gravitation: Every object exerts a gravitational pull on every other object. The force is proportional to the masses of both objects and inversely proportional to the square of the distance between their centers,

$$F \propto \frac{m_1 m_2}{d^2}$$

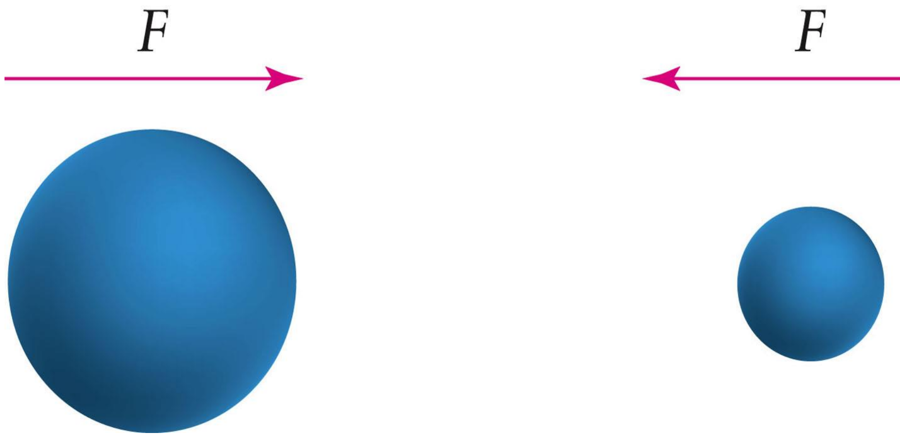
where m_1 and m_2 are the masses of the two objects, and d is the distance between their centers.

2.7 The Law of Universal Gravitation

- The equal and opposite gravitational forces that act on any pair of objects depend on the masses of both objects.

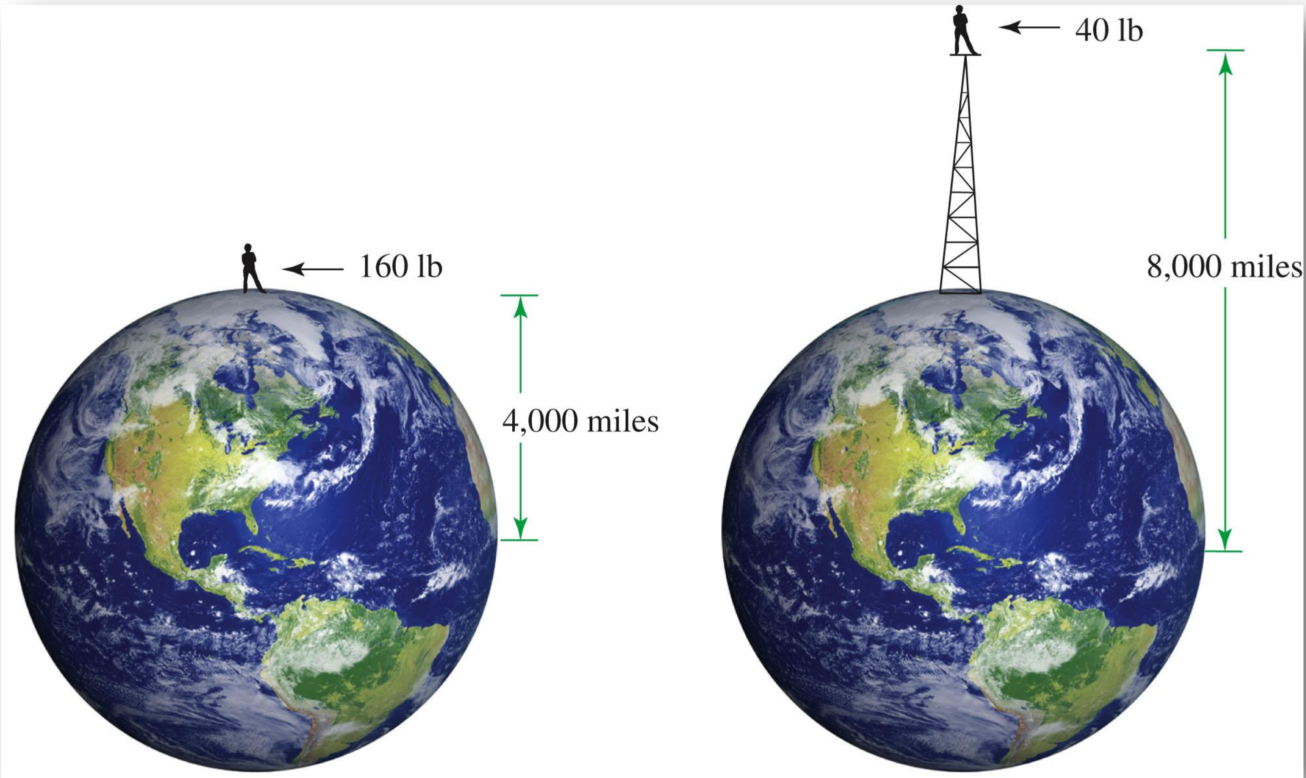


- If either object is replaced by another with a larger mass, the two forces are proportionally larger.



2.7 The Law of Universal Gravitation

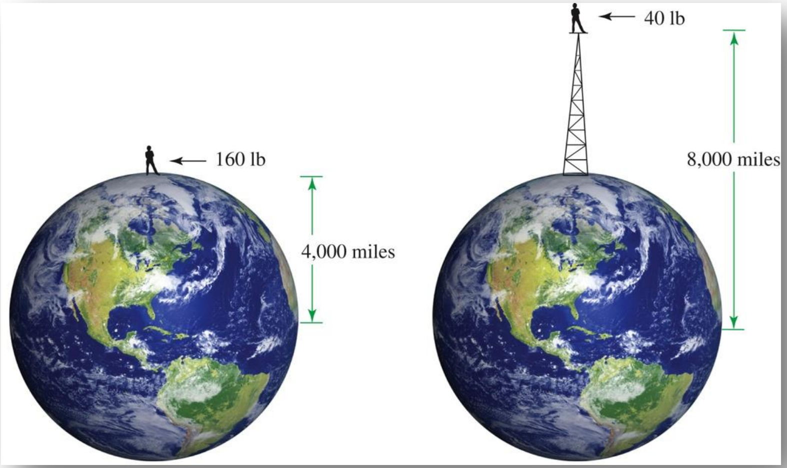
The size of the gravitational force between two bodies must also depend on the distance between them.



2.7 The Law of Universal Gravitation

- The Sun is much more massive than Earth, but its force on you is much less than Earth's, because you are much closer to Earth.

- If you were twice as far from Earth's center (8,000 miles instead of 4,000 miles), you would weigh one-fourth as much.



2.7 The Law of Universal Gravitation

Note that Newton's 3rd law means:

- the Earth pulls down on you with a force equal to your weight, and
- you pull "up" on the Earth with you're a force equal to your weight.

You move more than the Earth because the Earth is much more massive than you.

Example

A space probe is launched from the Earth, headed for deep space. At a distance of 10,000 miles from the Earth's center, the gravitational force on it is 600 lb. What is the size of the force when it is 20,000 miles from the Earth's center?

The problem gives us:

$$d_1 = 10{,}000 \text{ mi}$$

$$F_1 = 600 \text{ lb}$$

$$d_2 = 20{,}000 \text{ mi}$$

The force is:

$$F = \frac{GMm}{d^2}$$

Example

We can write these equations as

$$F_1 = \frac{GMm}{d_1^2} \quad \text{and} \quad F_2 = \frac{GMm}{d_2^2}$$

$$F_1 d_1^2 = GMm \quad \text{and} \quad F_2 d_2^2 = GMm$$

Example

we get $$F_1 d_1^2 = F_2 d_2^2$$

The requested force is

$$F_2 = F_1 \frac{d_1^2}{d_2^2}$$

$$= \left(600\ \text{lb}\right)\frac{\left(10{,}000\ \text{mi}\right)^2}{\left(20{,}000\ \text{mi}\right)^2} = \left(600\ \text{lb}\right)\times \frac{1}{4}$$

$$= 150\ \text{lb}$$

2.7 The Law of Universal Gravitation

The acceleration from gravity on the Moon is not the same as it is on Earth. Likewise, each of the other planets has its own "*g*." This variation exists simply because the masses and the radii are all different.

Location	Acceleration due to Gravity (*g*)
Earth	1.0
Sun	27.9
Moon	0.16
Mercury	0.38
Venus	0.88
Mars	0.39
Jupiter	2.65
Saturn	1.05

Important Equations

IMPORTANT EQUATIONS

Equation	Comments	Equation	Comments
Fundamental Equations		*Special-Case Equations*	
$F = ma$	Newton's second law of motion	$F = \frac{mv^2}{r}$	Centripetal force (object moving in a circular path)
$W = mg$	Relates weight and mass	$F = -kd$	Hooke's law
$F = \frac{Gm_1 m_2}{d^2}$	Law of universal gravitation		